TCET
DEPARTMENT OF VOCATIONAL STUDIES
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education
Under Autonomy - CBCGS-HME 2025
University of Mumbai

NAME: SAGARSUMAN P MANJHI **PRACTICAL: 08** **BATCH : 04**
SUBJECT: PYTHON PROGRAMMING LAB **SYBVOC :SEM-III (DS)**

EXPERIMENT BASED ON REGULAR EXPRESSIONS

AIM : To study and implement regular expressions in Python for pattern matching, validation, searching, splitting, and text replacement by developing a User Registration Form Validator for email, phone number, password, and PIN code.

LEARNING OBJECTIVE :

- To understand regular expressions in Python.
- To learn the use of the re module.
- To validate names, email addresses, phone numbers, passwords, and PIN codes.
- To use regular expression functions for pattern matching.
- To understand how to apply conditions for input validation.
- To develop a user registration validation program.

TOOLS :

Sr. No.	Name Of Resources	Specification	Quantity	Remarks
1.	Hardware	Computer (i5-i7) Ram (Min 16GB)	1	For All Practical
2.	Software	Python / IDLE / VS Code	1	For All Practical

THEORY :

Regular expressions, also known as regex, are sequences of characters used to define search patterns. Python provides the built-in re module to perform pattern matching, validation, searching, splitting, and text replacement.

In this practical, a User Registration Form Validator is developed to check the validity of a user's name, email address, phone number, password, and six-digit PIN code.

Concepts Used

- **Regular Expressions:** Patterns used to match specific text formats.
- **re Module:** A built-in Python module for working with regular expressions.
- **re.fullmatch():** Checks whether the entire string matches a specified pattern.
- **strip():** Removes leading and trailing whitespace from a string.
- **Conditional Statements:** The if, elif, and else statements determine whether the entered details are valid.
- **Raw Strings:** Patterns prefixed with r make it easier to write regular expressions containing backslashes.

Syntax

```
import re
if re.fullmatch(pattern, text):
  print("Valid")
else:
  print("Invalid")
```

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Source Code:

```python
# Practical 8: User Registration Validator
import re

name = input("Enter name: ")
email = input("Enter email: ")
phone = input("Enter phone: ")
password = input("Enter password: ")
pin = input("Enter 6-digit PIN: ")

if not name.strip():
    print("Invalid Name")
elif not re.fullmatch(r'[\w.-]+@[\w.-]+\.\w+', email):
    print("Invalid Email")
elif not re.fullmatch(r'[6-9]\d{9}', phone):
    print("Invalid Phone")
elif not re.fullmatch(r'\d{6}', pin):
    print("Invalid PIN")
elif not re.fullmatch(
    r'(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$!]).{8,}',
    password
):
    print("Invalid Password")
else:
    print("Registration Successful")
```

Output:

```
PS C:\Users\sagar\Documents\PYTHON> python Practicals/P8.py
Enter name: Sagarsuman Manjhi
Enter email: sagar@gmail.com
Enter phone: 2312435678
Enter password: 123ghy
Enter 6-digit PIN: 2323
Invalid Phone
```

Result and Discussion:

The User Registration Form Validator was implemented successfully using Python regular expressions and the re module.

The program validates the following details:

- **Name:** Checks that the name is not empty or composed entirely of whitespace.
- **Email:** Checks the basic email format, including an @ symbol and a domain suffix.
- **Phone Number:** Checks for a ten-digit Indian mobile number beginning with a digit from 6 to 9.
- **PIN Code:** Checks that the PIN contains exactly six digits.
- **Password:** Requires at least eight characters, one uppercase letter, one lowercase letter, one digit, and one special character from @, #, $, or !.

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Learning Outcomes:

- Learned to import and use the re module.
- Applied re.fullmatch() to validate input formats.
- Implemented validation for email, phone number, password, and PIN code.
- Used conditional statements to manage validation results.
- Developed a basic registration validation program in Python.

Course Outcomes:

- Understand regular expressions and their applications in Python.
- Apply pattern matching to validate user input.
- Use Python's built-in modules for text processing.
- Develop a registration validator using conditional statements and regular expressions.

CONCLUSION :

The User Registration Form Validator was developed successfully using Python regular expressions. The program validates the user's name, email address, phone number, password, and six-digit PIN code. This practical demonstrates how regular expressions can be used to perform input validation and improve the reliability of user registration programs.

VIVA QUESTIONS :

Q1. What is a regular expression?
A regular expression is a pattern used to search, match, or validate text.

Q2. Which module is used for regular expressions in Python?
The built-in re module is used.

Q3. What is the purpose of re.fullmatch()?
It checks whether the entire input string matches the specified pattern.

Q4. What does \d represent in a regular expression?
It matches a digit.

Q5. What does \d{6} mean?
It matches exactly six digits.

Q6. What is the purpose of [A-Z]?
It matches an uppercase English letter from A to Z.

FOR FACULTY USE ONLY :

Correction Parameters	Formative Assessment [40%]	Timely completion of practical [40%]	Attendance/ Learning Attitude [20%]	
Marks Obtained				